

Jefrey Bergl

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EDUCATION

University of North Carolina at Chapel Hill

Expected Graduation: May 2026

BS Computer Science and Applied Mathematics

Dean's List (5 semesters)

Relevant Coursework: Graduate Machine Learning, Graduate Computer Vision, Graduate Deep Learning, Optimization for Machine Learning, Numerical Analysis, Advanced Programming Language Concepts

RESEARCH EXPERIENCE

UNC Chapel Hill

Oct 2024 – Present

Undergraduate Research Assistant | Advisor: Prof. Thomas Hofweber

- Co-authored *The Black Tuesday Attack: How to Crash the Stock Market With Adversarial Attacks to Financial Forecasting Models* (submitted to Journal of Cybersecurity; see Publications).
- Developed adversarial perturbation framework for financial forecasting models; ran experiments on CNN and MLP architectures.
- Received \$3,000 in undergraduate research funding to support the project.

PUBLICATIONS & PREPRINTS

Thomas Hofweber, **Jefrey Bergl**, Ian Reyes, Amir Sadovnik. *The Black Tuesday Attack: How to Crash the Stock Market With Adversarial Attacks to Financial Forecasting Models*. Submitted to *The Journal of Cybersecurity*, 2025. [arXiv:2510.18990](https://arxiv.org/abs/2510.18990).

PROJECTS

H-JEPA: Learning Hamiltonian Dynamics on Latent Video Embeddings [[GitHub](#)]

Dec 2025

Independent Research | Python, PyTorch, V-JEPA 2

- Designed a latent world model that enforces Hamiltonian mechanics on frozen V-JEPA 2 video embeddings, learning a scalar energy function $H(q, p)$ and evolving states via a symplectic Störmer–Verlet integrator with a learnable timestep
- Achieved $1.9\times$ lower prediction MSE than a parameter-matched baseline MLP on 15-step rollouts, using 30% fewer parameters (919K vs. 1.3M), with bounded energy drift by construction
- Conducted ablation studies isolating contributions of symplectic integration and energy conservation loss, revealing a prediction–conservation trade-off across four model variants

The Representation-Reasoning Gap: Encoding vs. Reasoning in LMs [[GitHub](#)]

Jan 2026

Independent Research | Python, PyTorch, TransformerLens

- Demonstrated a dissociation between representation formation and reasoning in language models (GPT-2 Medium, Pythia 1.4B, Pythia 2.8B): linear probes recover spatial position with 98–100% accuracy from residual stream activations while behavioral accuracy is 0–5%
- Designed a controlled 5×5 grid-world navigation task with 2,500 scenarios and trained ridge classifiers on residual stream activations across all layers and step counts
- Discovered that position representations *consolidate* at the final token (up to +22pp over step tokens) yet models still fail, ruling out information loss and localizing the bottleneck to the unembedding layer

PRESENTATIONS

CoreAI UNC Research Showcase – “The Black Tuesday Attack” [[poster](#)]

Feb 2026

AI@UNC – “Adversarial Attacks on Financial Forecasting Models” [[slides](#)]

April 2025

Selected for Spotlight Presentation

COMP 790: Graduate Computer Vision – “Evaluating Real World ViT Performance” [[poster](#)] May 2025

TECHNICAL SKILLS

Programming Languages: Python, Java